

January 2026 CSE 406: Cyber Security Sessional

Online 1: Buffer Overflow

Section: A2

Time: 25 minutes

Objective

You are given a vulnerable C program (`target.c`) that contains a buffer overflow vulnerability. Your task is to **chain two function calls** using return address manipulation:

1. First, redirect to `unlock(code)` with the correct unlock code
2. Then, after `unlock()` finishes, redirect to `get_reward()` which spawns a shell

Environment Setup

```
seed@ubuntu:~$  
sudo sysctl -w kernel.randomize_va_space=0  
sudo ln -sf /bin/zsh /bin/sh
```

Student ID Configuration

Open `target.c` and replace `<LAST_3_DIGITS_OF_YOUR_ID>` in the `#define STUDENT_ID` line with the **last 3 digits** of your Student ID.

Important: The `UNLOCK_CODE` is derived from your Student ID: `0xA5A5A000 + STUDENT_ID`. Your exploit must pass this exact value.

Compilation

```
seed@ubuntu:~$  
gcc -m32 -o target -fno-stack-protector target.c  
sudo chown root target  
sudo chmod 4755 target  
touch badfile  
gcc -m32 -o target_dbg -g -fno-stack-protector target.c
```

Task

Write a Python exploit script (`exploit.py`) that generates a `badfile` such that when the program is run, it prints:

```
seed@ubuntu:~$  
Checking code: 0xa5a5aXXX  
Code accepted! Vault unlocked!  
Treasure claimed! You win!
```

And spawns a root shell.

Restrictions

1. **Do NOT** modify `target.c` (except for setting your `STUDENT_ID`)
2. You must call **both** `unlock()` **AND** `get_reward()` — calling only `get_reward()` is not sufficient
3. The `unlock()` function calls `exit(1)` if the wrong code is provided — you **MUST** pass the correct `UNLOCK_CODE`

Submission

1. Create a folder with your student ID as name (i.e., 2105185)
2. Put `target.c` and `exploit.py` inside it.
3. Zip the folder and upload it.